Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 10: LTI Resistive Circuits and Equivalent One-Ports

M. Mert Ankaralı

Colour code used in these notes and in the slides

current i voltage v a part of a circuit (box N), terminals a, b

Why this lecture? So far, KCL and KVL were written for any circuit, with any elements. From now on we restrict the elements. In an **LTI resistive circuit** every equation is linear and algebraic, so the circuit can be simplified before it is solved. This lecture gives the tools for that: the substitution theorem, one-ports and their equivalents, series and parallel connections, and the exchange of a voltage source and a current source.

1. LTI resistive circuits

Three pairs of words describe a circuit. Each pair is defined by what the circuit contains.

Nonlinear, time-varying, dynamic

- A **nonlinear** circuit contains a nonlinear resistor, inductor, capacitor or dependent source. A circuit is **linear** if it is not nonlinear.
- A **time-varying** circuit contains a time-varying resistor, inductor, capacitor or dependent source. A circuit is **time-invariant** if it is not time-varying.
- A **dynamic** circuit contains an inductor or a capacitor. A circuit is **resistive** if it is not dynamic.

An **LTI resistive circuit** is linear, time-invariant and resistive. It contains only LTI resistors, linear time-invariant dependent sources and independent sources.

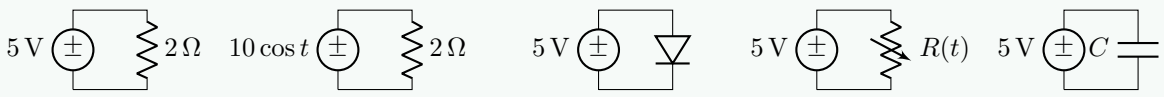
Independent sources do not count

The lists above do not mention independent sources. A source $v_s(t) = 10\cos t\,\mathrm{V}$ driving an LTI resistor gives an LTI resistive circuit. The words describe the circuit, not the waveform that drives it.

What we gain. In an LTI resistive circuit KCL and KVL are linear, and every element equation is linear and algebraic: $v = Ri$ for a resistor, $v = \mu v_x$ or $i = g\,v_x$ for a dependent source. No derivative appears. Time enters only through the values of the independent sources, so at each instant the circuit is solved like a circuit with constant sources.

Example 10.1 — which circuit is which?

Each circuit is a source and one more element.

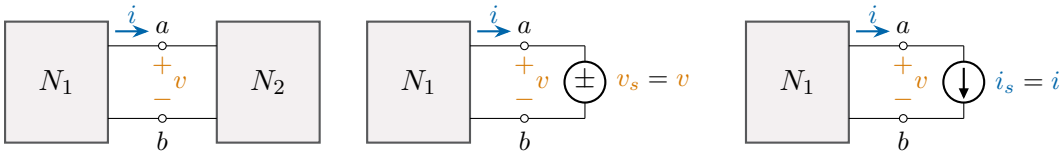


(a) LTI resistive (b) LTI resistive (c) nonlinear (diode) (d) time-varying (e) dynamic

(b) is LTI resistive although its source varies in time. In (d), $R(t) = 2 + \sin t\ \Omega$. (c) and (d) are still resistive; (e) is still linear and time-invariant.

2. The substitution theorem

Split a circuit into two parts N_1 and N_2, joined at two terminals a and b. Let $v(t)$ be the voltage between a and b, and $i(t)$ the current from N_1 into N_2 at a.



(I) the circuit (II) $N_2 \to$ voltage source (III) $N_2 \to$ current source

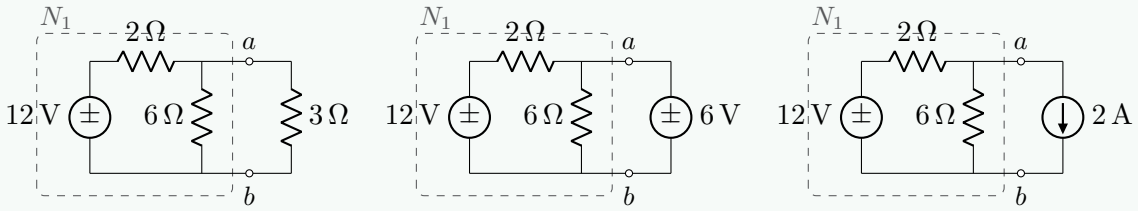
Substitution theorem

Replace N_2 by an independent voltage source $v_s(t) = v(t)$, or by an independent current source $i_s(t) = i(t)$. No branch voltage or branch current inside N_1 changes. An observer inside N_1 cannot tell (I), (II) and (III) apart.

Why it works. The equations inside N_1 (its KCL, KVL and element equations) do not change. N_1 meets the rest of the circuit only through v and i. The source fixes one of them at the value it had in (I), and the solution of (I) still satisfies every equation of (II) and (III). N_2 can be anything, even nonlinear: only its port values matter.

One condition. The new circuit must have a unique solution. If N_1 is a single current source, replacing N_2 by a current source puts two current sources in series: their currents are equal, but the voltage v is no longer fixed.

Example 10.2 — substitution in a small circuit



(I) N_2: the 3 Ω resistor (II) a 6 V source (III) a 2 A source

(I) The 6 Ω and 3 Ω are in parallel: 2 Ω. The two 2 Ω share the 12 V, so $v = 6$ V, and $i = 6/3 = 2$ A flows into N_2. Inside N_1: the 2 Ω carries $(12-6)/2 = 3$ A and the 6 Ω carries $6/6 = 1$ A.

(II) The 6 V source fixes $v = 6$ V. Again the 2 Ω carries 3 A and the 6 Ω carries 1 A; the source takes the remaining 2 A.

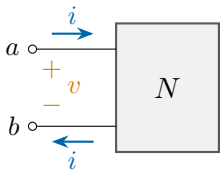
(III) The 2 A source takes 2 A from node a. KCL at a: $(12-v)/2 = v/6 + 2$, so $v = 6$ V, and the currents inside N_1 are 3 A and 1 A again.

The theorem lets us cut a circuit and study one part at a time. It is also the key step in later proofs, where a complicated part is replaced by a source.

3. Ports and equivalent one-ports

Port, one-port

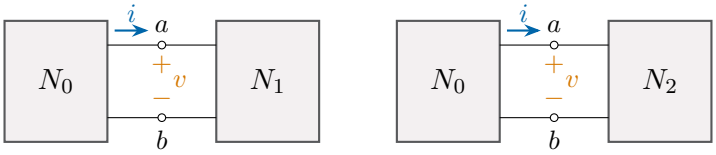
A pair of terminals is a **port** if the current entering one terminal equals the current leaving the other. A **one-port** is a circuit, or a circuit element, with one pair of terminals accessible from outside.



A one-port is always a port: by KCL on a closed surface around N, the current i that enters at a leaves at b. We take i entering the + terminal of v. The relation between v and i is the i–v **characteristic** of the one-port. A resistor is the simplest one-port: $v = Ri$.

Equivalent one-ports

Two one-ports N_1 and N_2 are **equivalent**, $N_1 \equiv N_2$, if they have the same i–v characteristic at their terminals. Then, for any one-port N_0 connected to them, every branch voltage and branch current inside N_0 is the same with N_1 or with N_2.

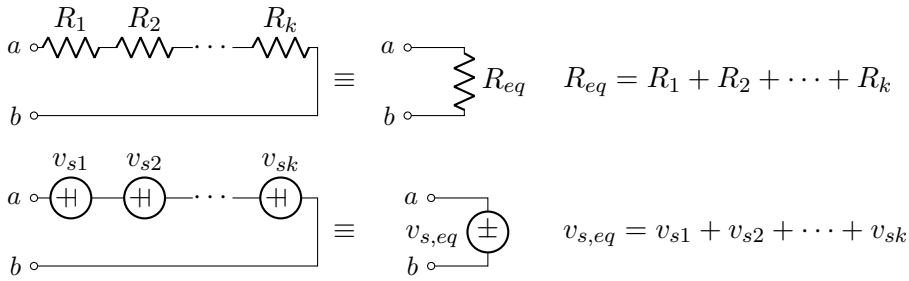


N_0 meets the rest only through v and i. The equations of the whole circuit are those of N_0 plus the i–v characteristic at the port. Equivalent one-ports give the same equations, so the same solution inside N_0. Inside N_1 and N_2 the branch values may differ.

4. Series and parallel connections

Series connection

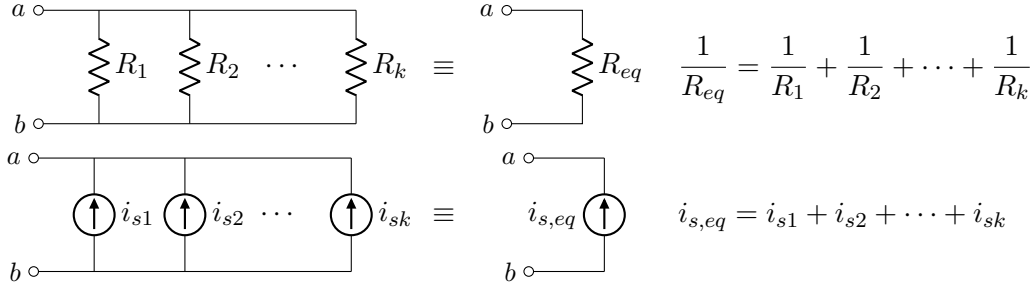
In a series connection the same current i flows through every element (KCL), and the voltages add (KVL).



For the resistors: $v = R_1 i + \cdots + R_k i = R_{eq} i$. The share of R_j is $v_j = (R_j/R_{eq})\, v$ (**voltage divider**). For the sources all + signs point towards a; a source turned the other way enters the sum with −.

Parallel connection

In a parallel connection every element has the same voltage v (KVL), and the currents add (KCL).



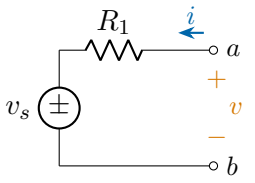
With conductances $G_j = 1/R_j$: $i = G_1 v + \cdots + G_k v = G_{eq} v$, so $G_{eq} = G_1 + \cdots + G_k$. The share of R_j is $i_j = (G_j/G_{eq})\, i$ (**current divider**). Two resistors: $R_{eq} = R_1 R_2/(R_1 + R_2)$.

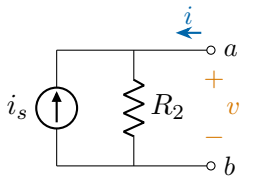
Two connections that are not allowed

Two voltage sources with $v_{s1} \neq v_{s2}$ in parallel break KVL. Two current sources with $i_{s1} \neq i_{s2}$ in series break KCL. With equal values the connection is allowed, but the current shared by the two voltage sources (or the voltage shared by the two current sources) is not fixed.

5. One source and one resistor

Take a voltage source with a series resistor, and a current source with a parallel resistor. In both, i enters at a.





$$v = R_1 i + v_s \qquad (1) \qquad\qquad v = R_2(i + i_s) = R_2 i + R_2 i_s \qquad (2)$$

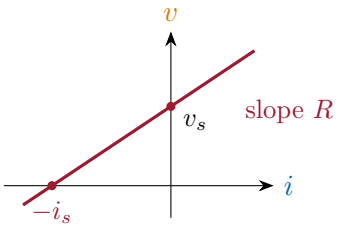
In (2), i and i_s both flow into the top node, so R_2 carries $i + i_s$. Both characteristics are straight lines in the i–v plane. They are the same line when the slopes and the intercepts agree.

Source transformation

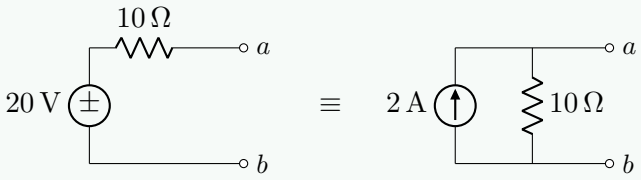
The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$. Short circuit ($v = 0$): $i = -i_s$.



Example 10.3 — a source transformation



$R = 10\,\Omega$ and $i_s = v_s/R = 20/10 = 2\,\text{A}$. Both have open-circuit voltage $20\,\text{V}$ and short-circuit current $2\,\text{A}$ (out of a).

Example 10.4 — Example 10.2 again, by transformations

Replace the $12\,\text{V}$ source and the $2\,\Omega$ by a $6\,\text{A}$ source in parallel with $2\,\Omega$. Now $2\,\Omega$, $6\,\Omega$ and $3\,\Omega$ are in parallel: $G_{eq} = \frac{1}{2} + \frac{1}{6} + \frac{1}{3} = 1\,\text{S}$, so $R_{eq} = 1\,\Omega$, and

$$v = 6\,\text{A} \times 1\,\Omega = 6\,\text{V}, \qquad i = 6/3 = 2\,\text{A},$$

as before. The part outside the transformed one-port sees the same v and i. Inside it the values differ: the $12\,\text{V}$ source carries $3\,\text{A}$, while the $6\,\text{A}$ source carries $6\,\text{A}$.

6. Solving LTI resistive circuits

Series and parallel connections and source transformations simplify many circuits, but not all. Two systematic methods solve every LTI resistive circuit:

1. **node voltage analysis** — the $n-1$ node voltages are the unknowns;
2. **mesh current analysis** — the $b-n+1$ mesh currents are the unknowns.

Both start from the matrices met earlier (A and M); now the element equations are linear and can be written in matrix form too.

Summary

- A circuit is **linear**, **time-invariant** and **resistive** if it has no nonlinear element, no time-varying element and no L or C. Independent sources do not count. All equations of an LTI resistive circuit are linear and algebraic.
- **Substitution:** a part N_2 may be replaced by a voltage source equal to its port voltage, or a current source equal to its port current; nothing changes in the rest (if the solution stays unique).
- Two one-ports are **equivalent** if they have the same i–v characteristic; the rest of the circuit cannot tell them apart.
- Series: $R_{eq} = \sum R_j$, $v_{s,eq} = \sum v_{sj}$. Parallel: $G_{eq} = \sum G_j$, $i_{s,eq} = \sum i_{sj}$.
- A voltage source v_s with series R is equivalent to a current source $i_s = v_s/R$ with parallel R.